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Profile

Full-stack engineer specialising in backend systems and API design. I build production applications with Laravel and React from secure authentication flows to async AI pipelines and take products from requirements through UI design with the user experience front of mind. I dig into details, stress-test interfaces as a real user rather than just a builder, and communicate clearly across every stage of a project. Currently studying Computer Science at Addis Ababa University while shipping real projects used in live environments.

Technical Skills

Languages	PHP, JavaScript, TypeScript, Python, C++
Frontend	React, Next.js, Tailwind CSS, Vite, TanStack Query
Backend	Laravel, REST API Design, Sanctum, Node.js
Database	PostgreSQL, MySQL, Supabase, pgvector, GIN Indexing
Infrastructure	Docker, Redis, GitHub Actions, Linux, Vercel, Render

Projects

ZeroDay AI-Powered Security Intelligence Platform

🔗 Live | 🐙 GitHub

Laravel 13, React, PostgreSQL, Redis, Docker, Google Gemini AI

- Replaced standard SQL queries with a hybrid GIN + pgvector search engine, achieving sub-millisecond latency for both keyword and semantic search.
- Offloaded LLM processing to Redis background workers, keeping the UI responsive during AI triage with no blocking on the main thread.
- Secured the platform with HttpOnly session auth, Laravel Policy RBAC, and tiered rate limiting deployed and running in a live environment.
- Designed the platform's information architecture and UI flows by stress-testing each screen as an end user, ensuring security intelligence surfaced was actionable, not just present.

ACT E-Learning Management Platform

🐙 GitHub

React, Laravel, PostgreSQL, Sanctum

- Led a team through the full product lifecycle from requirements gathering and system design to UI prototyping and final delivery coordinating decisions and communication across all project phases.
- Built a multi-role LMS supporting video lessons, dynamic quizzes, and progress tracking across 3 user roles with scoped access at every layer.
- Eliminated write conflicts under concurrent quiz submissions using PostgreSQL atomic transactions, ensuring consistent scoring at scale.
- Authenticated all API endpoints with Sanctum, separating student and admin sessions with distinct permission scopes.

Le-Restaurant Real-Time Restaurant POS System

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Next.js, TypeScript, Supabase, Tailwind CSS

- Replaced polling with Supabase WebSocket subscriptions for real-time order sync across admin and client devices.
- Optimised rendering and data fetching to score 95+ on Lighthouse in production.

Education

Addis Ababa University
B.Sc. in Computer Science

Addis Ababa, Ethiopia
Sep 2023 – May 2027 (Expected)

Certifications

ProDev Frontend Engineer ALX Software Engineering Programme

🔗 View

Oct 2025